

Recovery of San Francisco Cultural Institutions After COVID-19

Summary

After observing and exploring annual performance data reported by San Francisco departments, I discovered that the years of the COVID-19 pandemic had a massive effect on many of San Francisco's institutions; particularly, the cultural institutions. Which makes sense, as the pandemic dramatically disrupted public life, and institutions that rely on in-person participation suffered greatly. Museums, libraries, and cultural centers experienced prolonged closures and severe reductions in attendance. This project measures the drop-off in attendance for several prominent institutions, and investigates whether major San Francisco cultural institutions have recovered to their pre-pandemic engagement levels.

I analyzed participation trends for five major institutions: the San Francisco Public Library, the de Young Museum, the Legion of Honor, the Asian Art Museum, and the California Academy of Sciences. Note: while the War Memorial is an important SF cultural institution, the missingness in their reported data made analysis impossible. Recovery was measured by comparing pre-pandemic visitor levels to post-pandemic visitor levels and by examining changes in participation trends over time.

The analysis finds that although all institutions experienced substantial declines during the pandemic, recovery has been uneven. The Legion of Honor has nearly returned to pre-pandemic participation levels, while the Public Library and Asian Art Museum remain substantially below their historical baselines. A pooled interrupted time series model suggests that participation was already declining slightly before COVID-19, that the pandemic caused an immediate and severe reduction in attendance, and that participation has generally increased since reopening.

Overall, the results indicate that San Francisco's cultural institutions are recovering, but most have not yet fully returned to pre-pandemic participation levels.

Introduction

Cultural institutions play an important role in civic life by providing educational opportunities, preserving cultural heritage, and creating spaces for public engagement. Because these institutions depend heavily on in-person participation, they were especially vulnerable to the disruptions caused by the COVID-19 pandemic.

Museums and libraries experienced temporary closures, capacity restrictions, and shifts in visitor behavior that substantially reduced attendance. While many institutions have since reopened, it remains unclear whether participation has returned to historical levels.

This project mainly examines the question:

Have San Francisco cultural institutions recovered to their pre-pandemic participation levels?

Data

The analysis uses the San Francisco Citywide Performance Metrics dataset, which contains annual performance measures reported by city departments.

The original dataset contains:

- 51 departments:
 - ACADEMY OF SCIENCES
 - ADMINISTRATIVE SERVICES
 - ADULT PROBATION
 - AIRPORT COMMISSION
 - ARTS COMMISSION
 - ASIAN ART MUSEUM
 - ASSESSOR / RECORDER
 - BOARD OF APPEALS
 - BOARD OF SUPERVISORS
 - BUILDING INSPECTION
 - CHILD SUPPORT SERVICES
 - CHILDREN YOUTH & THEIR FAMILIES
 - CITY ATTORNEY
 - CITY PLANNING
 - CIVIL SERVICE COMMISSION
 - COMMUNITY INVESTMENT AND INFRASTRUCTURE
 - CONTROLLER
 - DISTRICT ATTORNEY
 - EARLY CHILDHOOD
 - ECONOMIC AND WORKFORCE DEVELOPMENT
 - ELECTIONS
 - EMERGENCY MANAGEMENT
 - ENVIRONMENT
 - ETHICS COMMISSION
 - FINE ARTS MUSEUM

- FIRE DEPARTMENT
- HEALTH SERVICE SYSTEM
- HOMELESSNESS AND SUPPORTIVE HOUSING
- HUMAN RESOURCES
- HUMAN RIGHTS COMMISSION
- HUMAN SERVICES
- JUVENILE PROBATION
- LAW LIBRARY
- MAYOR'S OFFICE OF HOUSING AND COMMUNITY DEVELOPMENT
- MUNICIPAL TRANSPORTATION AGENCY
- POLICE
- POLICE ACCOUNTABILITY
- PORT
- PUBLIC DEFENDER
- PUBLIC HEALTH
- PUBLIC LIBRARY
- PUBLIC UTILITIES COMMISSION
- PUBLIC WORKS
- RECREATION AND PARK COMMISSION
- RENT ARBITRATION BOARD
- RETIREMENT SYSTEM
- SHERIFF
- STATUS OF WOMEN
- TECHNOLOGY
- TREASURER/TAX COLLECTOR
- WAR MEMORIAL
-

- 13 variables:

- Department.Name
- Strategic.Goal
- Measure.Title
- Measure.Description
- Measure.Methodology
- Measure.Data.Type
- Fiscal.Year
- End.Calendar.Year
- Target
- Mid.Year.Result
- Mid.Year.Projection
- Full.Year.Result
- Data.Notes

The observations are annual, beginning between 2012 and 2015 depending on the institution, up through 2025. There are several missing observations resulting from incomplete reporting. Because the project focuses on cultural participation, only measures representing annual visitor counts were selected.

The final institutions included in the analysis are:

- San Francisco Public Library
- de Young Museum
- Legion of Honor
- Asian Art Museum
- California Academy of Sciences

Visitor counts were converted from character values to numeric values and cleaned before analysis.

Methodology

Baseline

To determine what participation looked like before the pandemic, I calculated average annual participation for each institution during the pre-pandemic period (2015–2019). Pre-pandemic growth rates were also estimated using simple linear regressions of annual visitor counts against time. These measures provide a benchmark against which pandemic and post-pandemic participation can be compared.

Measuring the Pandemic Shock

The pandemic period was defined as 2020–2021. For each institution, average visitor participation during these years was calculated and compared to the pre-pandemic baseline. This comparison provides a measure of the immediate disruption caused by COVID-19.

Measuring Recovery

Recovery was evaluated using a recovery ratio:

Recovery Ratio = Post-COVID Mean Participation / Pre-COVID Mean Participation

Interpretation:

- 1.00 = Full recovery
- Greater than 1.00 = Participation exceeds pre-pandemic levels
- Less than 1.00 = Participation remains below baseline

Post-COVID participation was defined using observations from 2022 onward.

Interrupted Time Series Analysis

To examine overall trends while controlling for differences in institution size, a pooled interrupted time series model was estimated:

$$\text{Participation} = \beta_0 + \beta_1(\text{Time}) + \beta_2(\text{Post-COVID}) + \beta_3(\text{Time} \times \text{Post-COVID}) + \text{Institution} + \varepsilon$$

The model estimates:

1. Pre-pandemic participation trends (β_1)
2. The immediate impact of COVID-19 (β_2)
3. The rate of recovery following reopening (β_3)

Institution fixed effects were included to account for differences in baseline attendance levels across organizations.

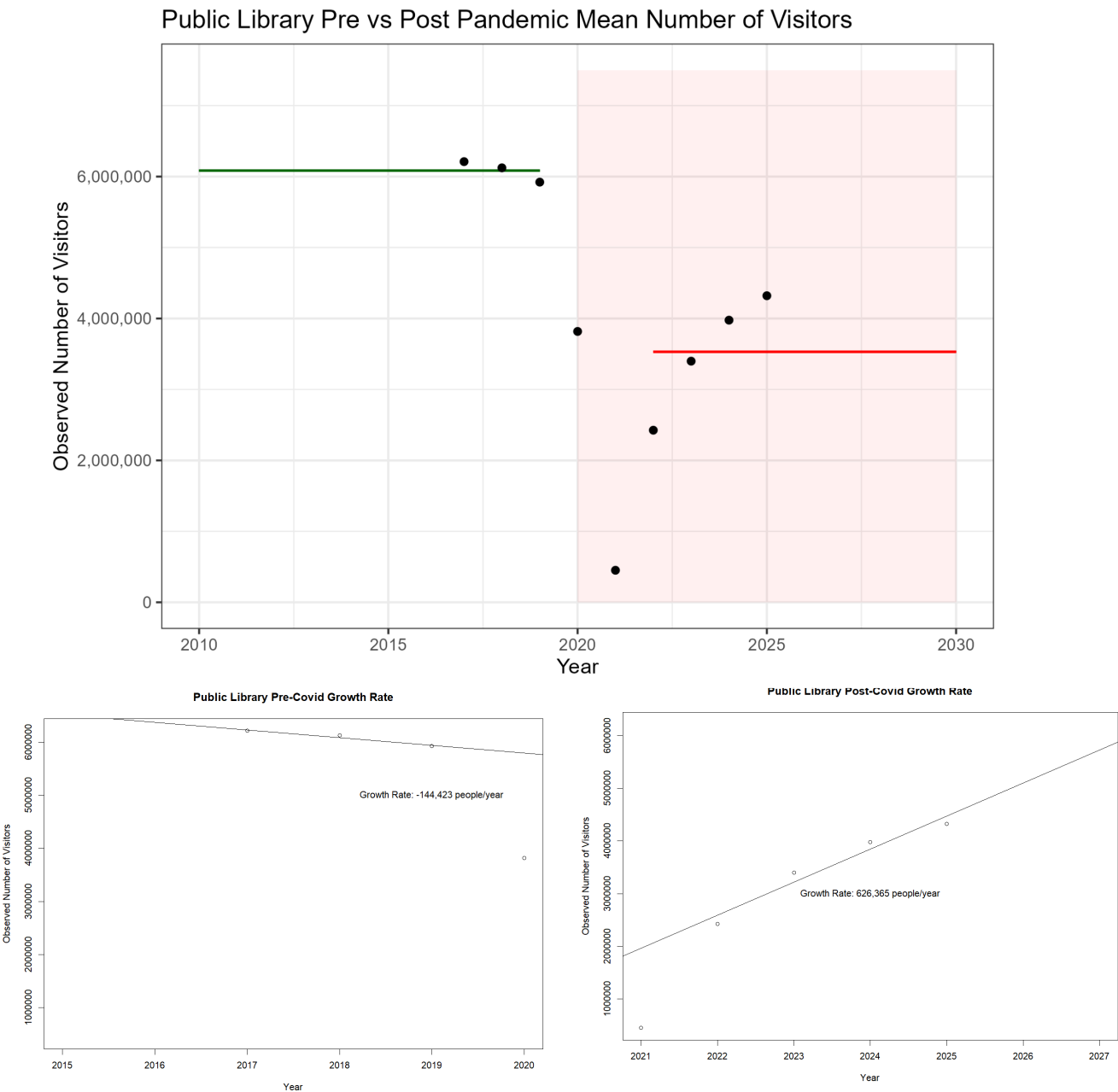
Results

Public Library

The San Francisco Public Library experienced one of the largest participation declines among the institutions examined.

- Pre-pandemic mean: 6.09 million visitors
- During the pandemic participation levels drop 154% to 2.14 million visitors
- Post-pandemic mean: 3.53 million visitors
- Recovery ratio: 0.58

Although attendance remains well below historical levels, post-pandemic growth rates indicate continued recovery



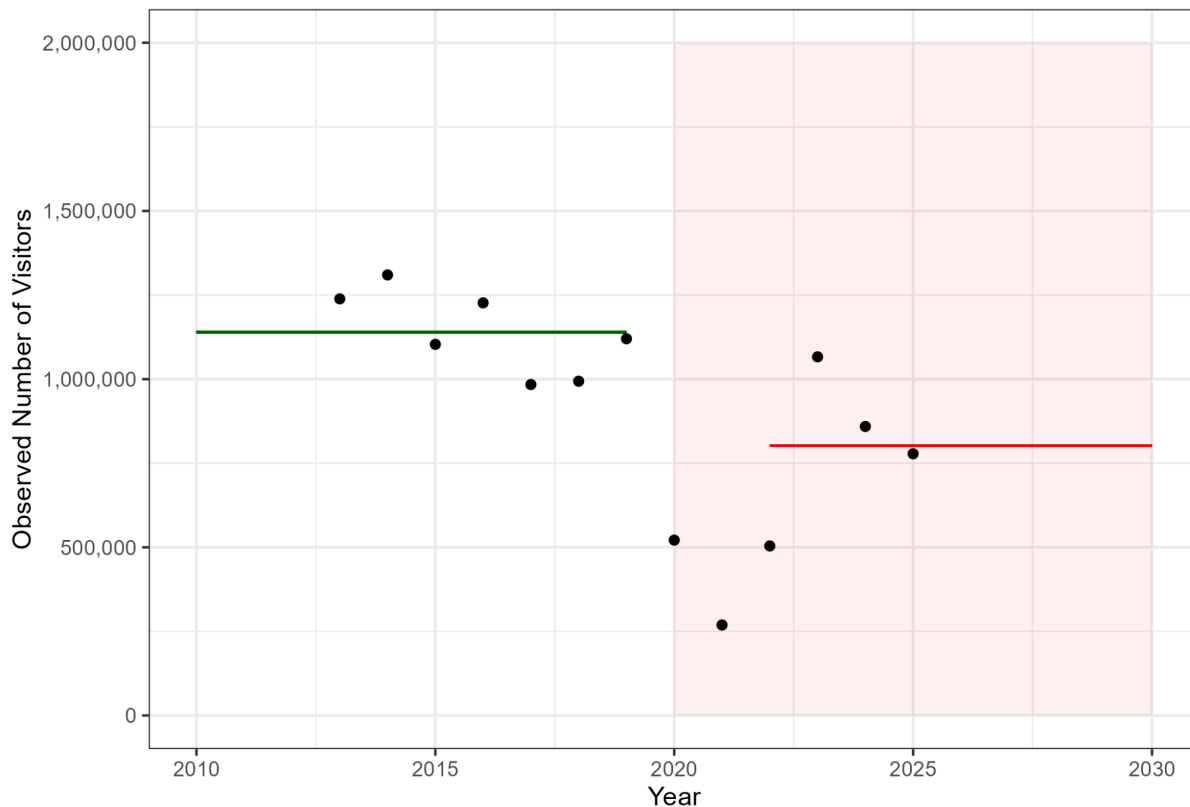
de Young Museum

The de Young Museum experienced substantial losses during the pandemic but has recovered more strongly than the Public Library.

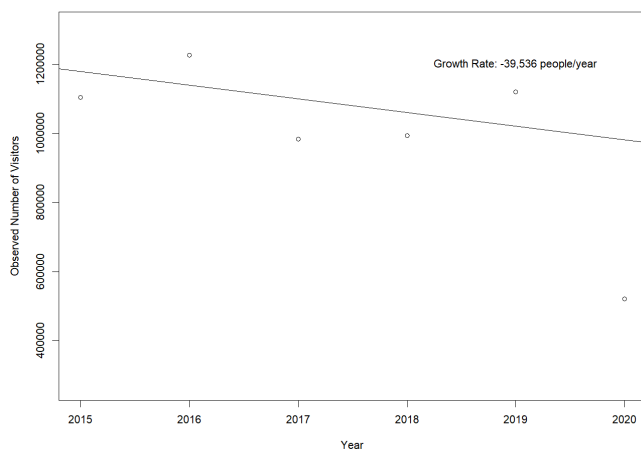
- Pre-pandemic mean: 1.14 million visitors
- During the pandemic participation levels dropped 153% to 395,225 visitors
- Post-pandemic mean: 802,000 visitors
- Recovery ratio: 0.70

Participation remains below baseline but shows evidence of sustained growth.

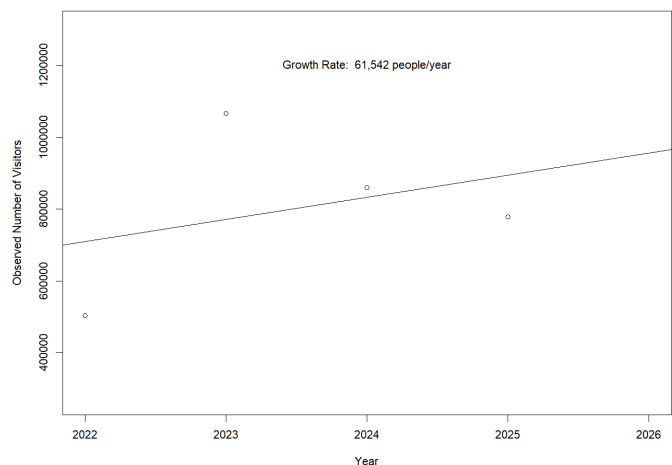
DeYoung Museum Pre vs Post 2020 Mean Number of Visitors



DeYoung Museum Pre-Covid Growth Rate



DeYoung Museum Post-Covid Growth Rate



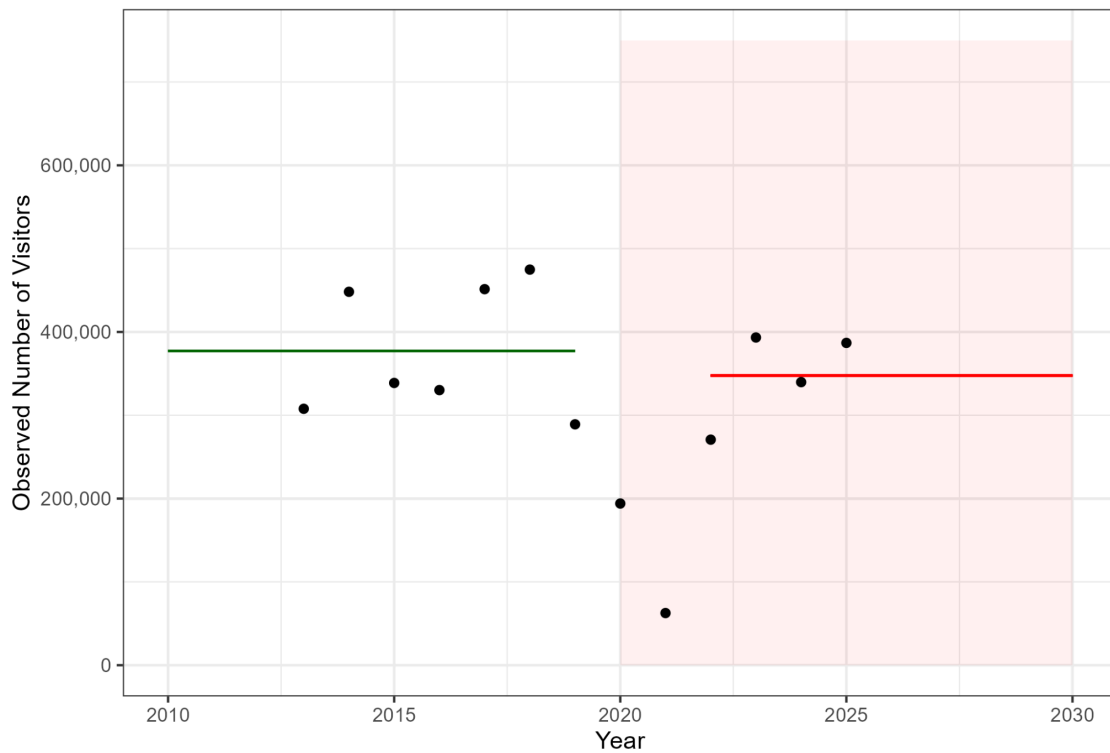
Legion of Honor

The Legion of Honor shows the strongest recovery among the institutions analyzed.

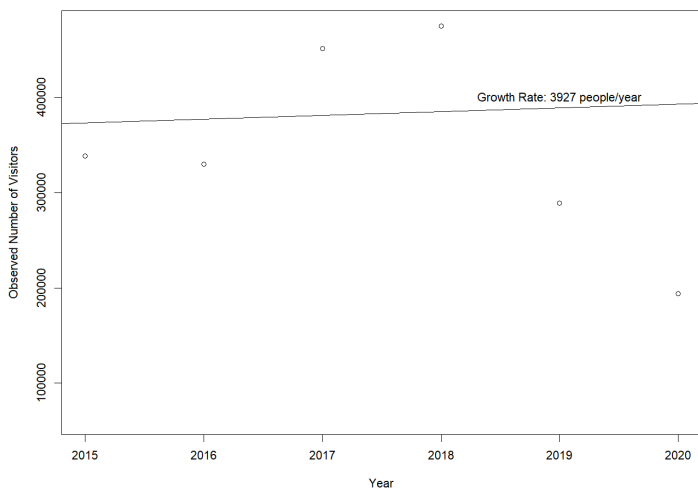
- Pre-pandemic mean: 377,198 visitors
- During the pandemic participation levels dropped 152% to 128,343 visitors
- Post-pandemic mean: 347,683 visitors
- Recovery ratio: 0.92

The museum has nearly returned to pre-pandemic attendance levels.

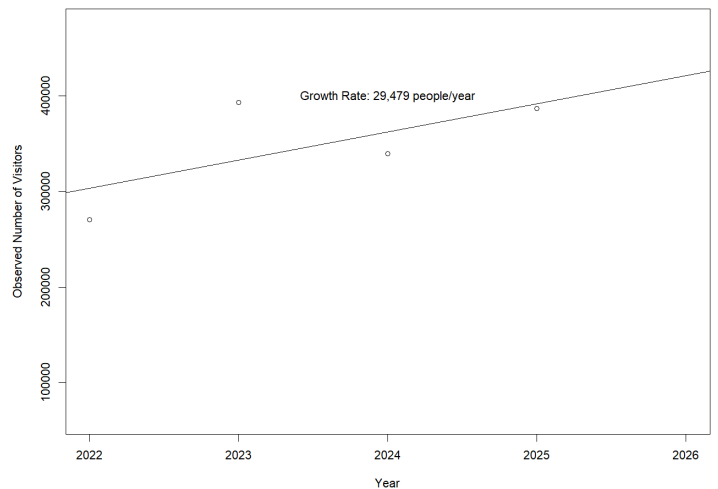
Legion Of Honor Museum Pre vs Post 2020 Mean Number of Visitors



Legion of Honor Pre-Covid Growth Rate



Legion Of Honor Post-Covid Growth Rate



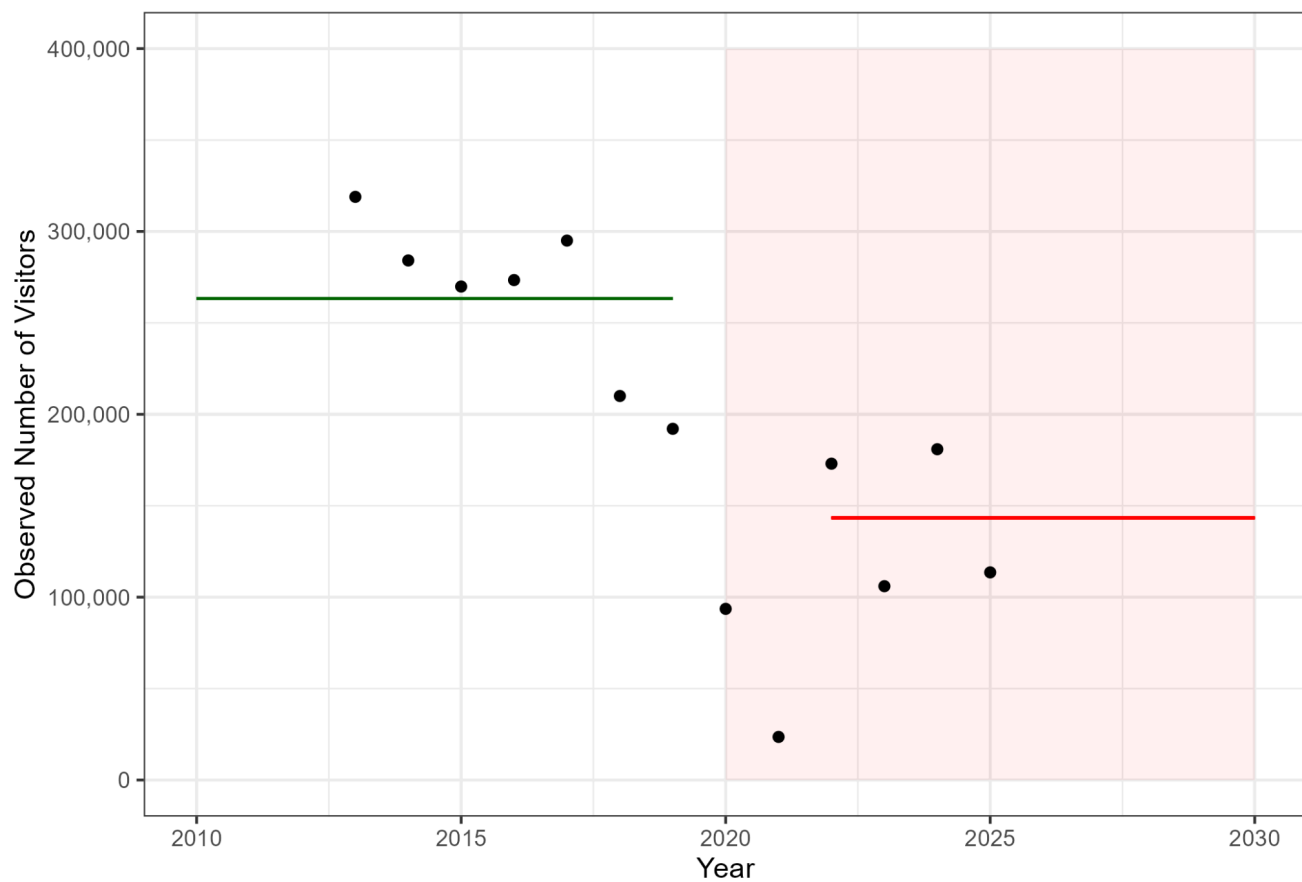
Asian Art Museum

The Asian Art Museum experienced both a severe pandemic shock and a slower recovery.

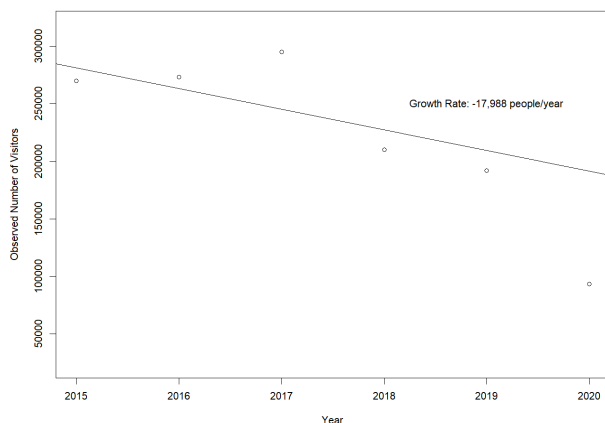
- Pre-pandemic mean: 263,343 visitors
- During the pandemic participation levels dropped 129% to 59,591
- Post-pandemic mean: 143,363 visitors
- Recovery ratio: 0.54

This represents the lowest recovery ratio among the institutions studied.

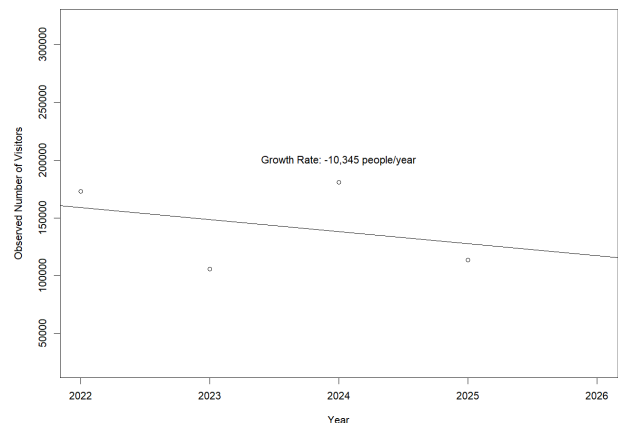
Asian Art Museum Pre vs Post 2020 Mean Number of Visitors



Asian Art Museum Pre-Covid Growth Rate



Asian Art Museum Post-Covid Growth Rate



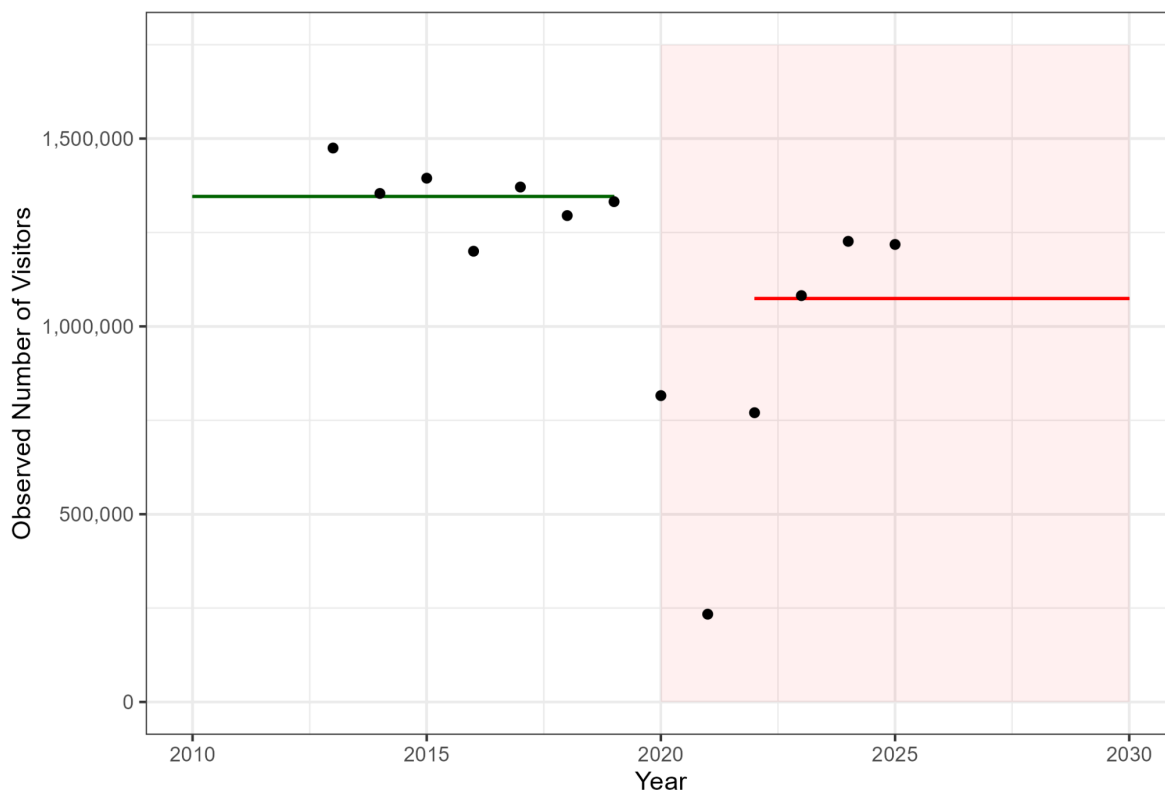
California Academy of Sciences

The California Academy of Sciences demonstrated a relatively strong rebound following pandemic closures.

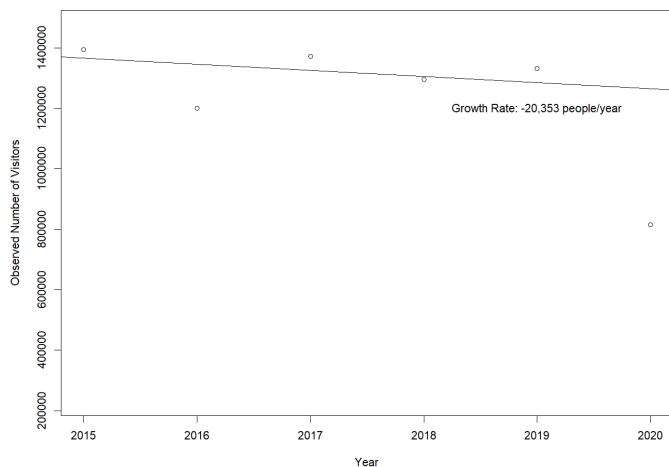
- Pre-pandemic mean: 1.35 million visitors
- During the pandemic, participation levels dropped 164% to 524,717
- Post-pandemic mean: 1.07 million visitors
- Recovery ratio: 0.80

Attendance remains below baseline but is substantially closer to full recovery than several other institutions.

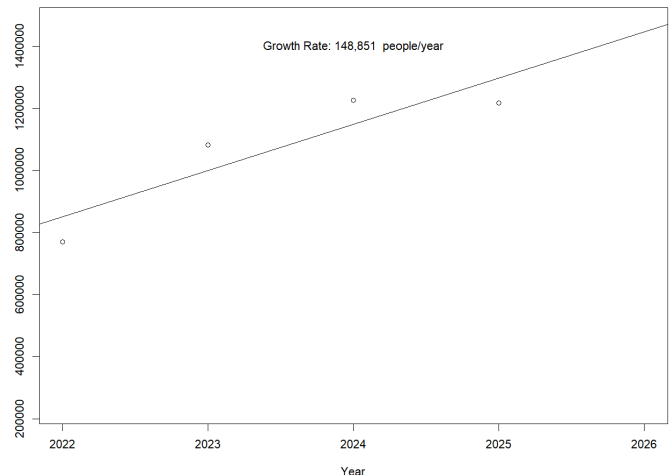
Academy of Science Pre vs Post 2020 Mean Number of Visitors



Academy Of Science Pre-Covid Growth Rate

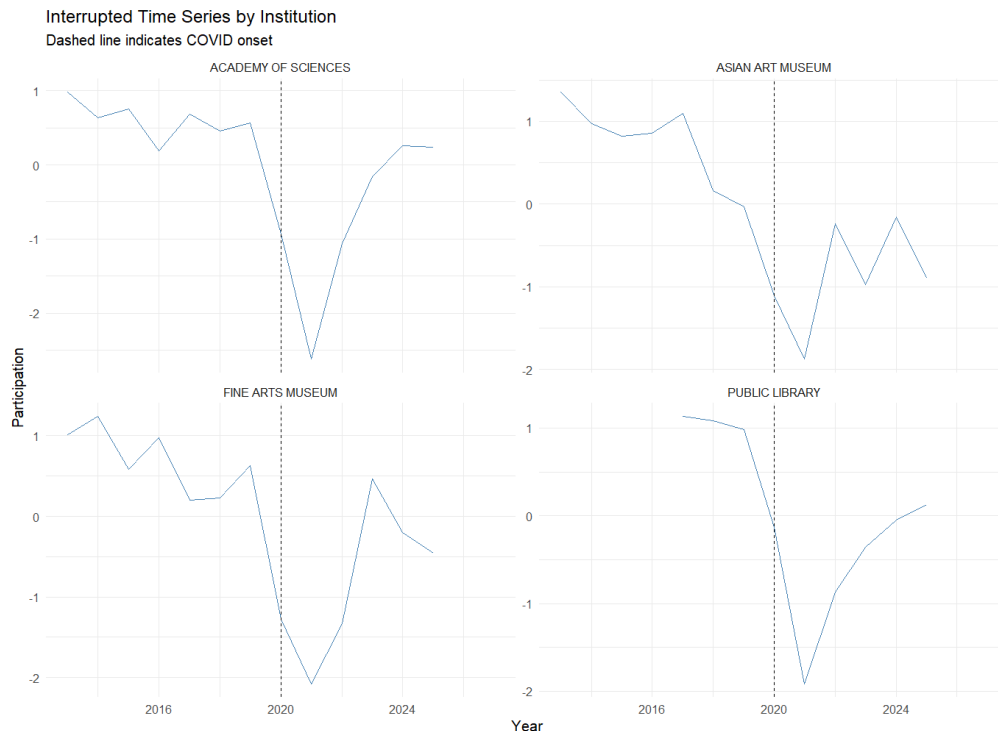


Academy Of Science Post-Covid Growth Rate



Interrupted Time Series Findings

The pooled interrupted time series model provides evidence of three important patterns. First, participation was already declining slightly before the pandemic. The estimated pre-COVID trend suggests an average decrease of approximately -105,176 visitors per year. Second, COVID-19 produced a large and immediate reduction in participation. The model estimates an average attendance loss of more than 750,000 visitors following the onset of the pandemic. Third, participation trends became positive after reopening. The estimated recovery trend suggests an increase of approximately 139,000 visitors per year following the pandemic period. Together, these findings suggest that while attendance has not yet fully recovered, institutions are generally moving in a positive direction.



Limitations

Several limitations should be considered when interpreting the results. The dataset contains annual observations, limiting the number of available data points and preventing more detailed time series analysis. Some institutions also contain missing observations resulting from inconsistent reporting practices. This limited the available data even further. Additionally, the interrupted time series model assumes that post-pandemic recovery follows a linear trend. In reality, recovery may be nonlinear and may differ substantially across

institutions. Because of these limitations, the results should be interpreted as descriptive rather than causal, and the numerical results should be treated as general estimates.

Conclusion

The COVID-19 pandemic produced a severe disruption in participation across San Francisco's cultural institutions. Every institution analyzed experienced substantial attendance declines during the pandemic period. Recovery has occurred, but it has been uneven. The Legion of Honor has nearly returned to pre-pandemic participation levels, while the Public Library and Asian Art Museum remain significantly below their historical baselines. The California Academy of Sciences and the de Young Museum fall between these extremes. These differences in recovery could be due to a number of reasons, from redesigning layouts to include more social distancing, restructuring of educational programs, or expansion of digital offerings. The interrupted time series analysis suggests that cultural participation is generally trending upward following reopening, indicating continued recovery. However, most institutions have not yet fully regained the visitor levels observed before COVID-19.